



# Big Data Visual Analytics (CS 661)

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# CS661 Quiz

- Date: July 6<sup>th</sup> Monday
- Where: LH-20
- When: 4:00pm-6:00pm
- Syllabus: Starting from lecture 11 (High Dimensional Data) - everything discussed until July 5th

# Study Materials for Lecture 16

- CoDDA: A Flexible Copula-based Distribution Driven Analysis Framework for Large-Scale Multivariate Data, Hazarika et al., IEEE TVCG.

# Multivariate Data Models using Distributions

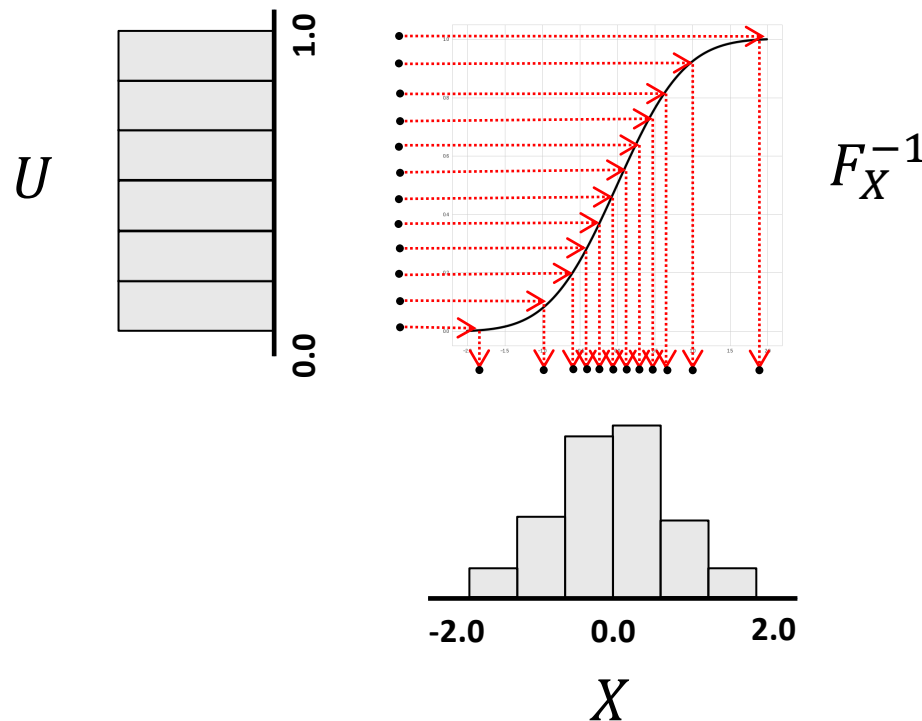
- Multivariate Histogram
  - Storage footprint of a multivariate histogram can increase exponentially with the number of variables and the desired level of discretization
  - Sparse representations can be used by storing only non-zero bins
- Multivariate GMM
  - Estimation of multivariate GMM using Expectation-Maximization is computationally expensive
  - Model complexity increases with the number of variables
- Multivariate KDE
  - Often computationally expensive and storage inefficient
- Important to preserve the dependency/correlations between variables for downstream multivariate analysis

# Concept of Copula

- **Definition:** A  $d$ -dimensional Copula is a CDF with uniform marginals. For  $d$ -uniform random variables, it can be denoted as  $\mathcal{C}(\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_d)$ .
- “Every joint CDF in  $\mathcal{R}^d$  inherently embodies a copula function”: Sklar (1959)
- Sklar’s Theorem provides the mathematical basis for splitting the problem of joint distribution estimation into two parts (using Copula functions):
  1. Estimating individual univariate marginal distributions.
  2. Estimating the dependencies between the random variables.

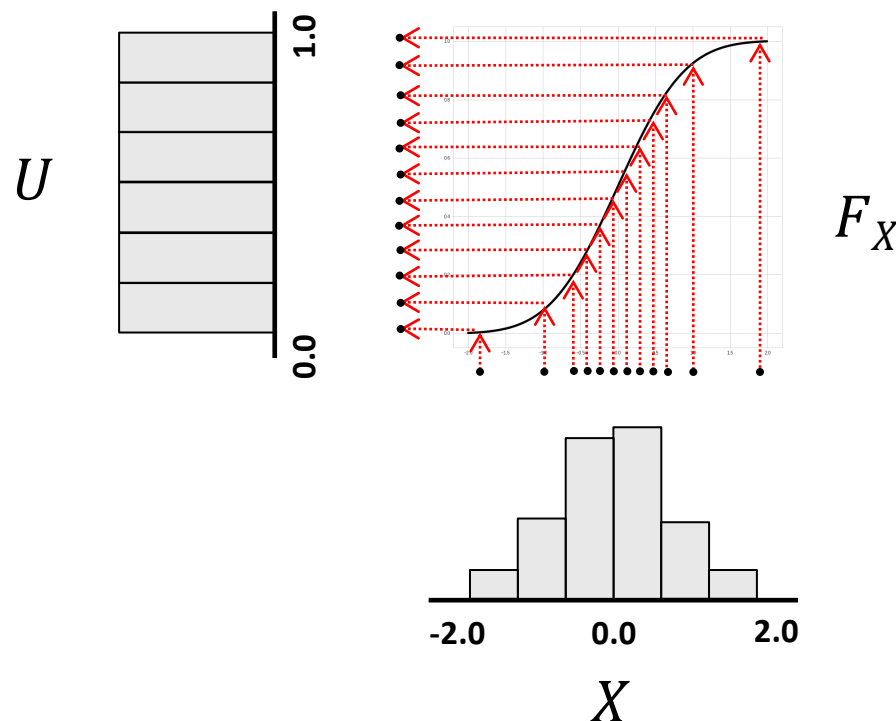
# Distribution Transformation Property

- **Property 1:** If  $U$  is a uniform random variable (i.e.,  $U \sim \text{Unif}[0,1]$ ) and  $F_X$  is a CDF of random variable  $X$ , then its inverse function  $F_X^{-1}(U)$  corresponds to the random variable  $X$  (i.e.,  $F_X^{-1}(U) \sim X$ )



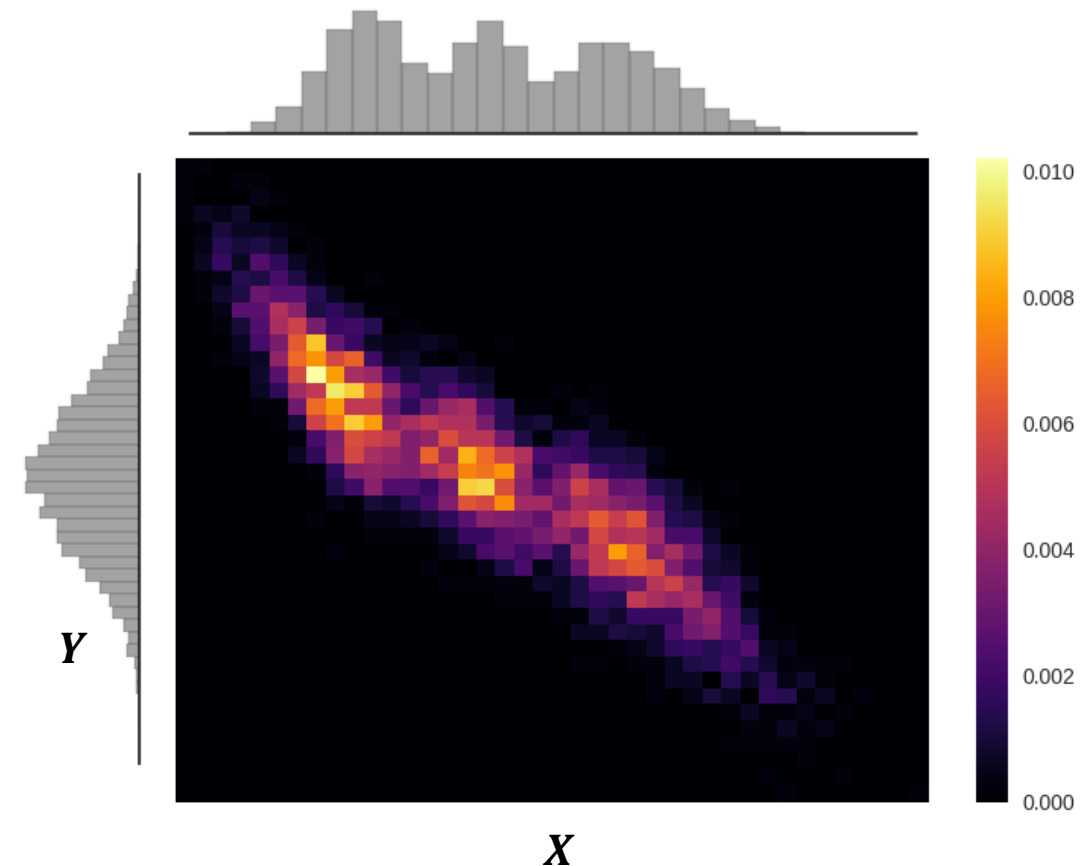
# Distribution Transformation Property

- **Property 2:** If a real-valued random variable  $X$  has a continuous cumulative distribution function  $F_X$ , then  $F_X(X) \sim Unif[0,1]$



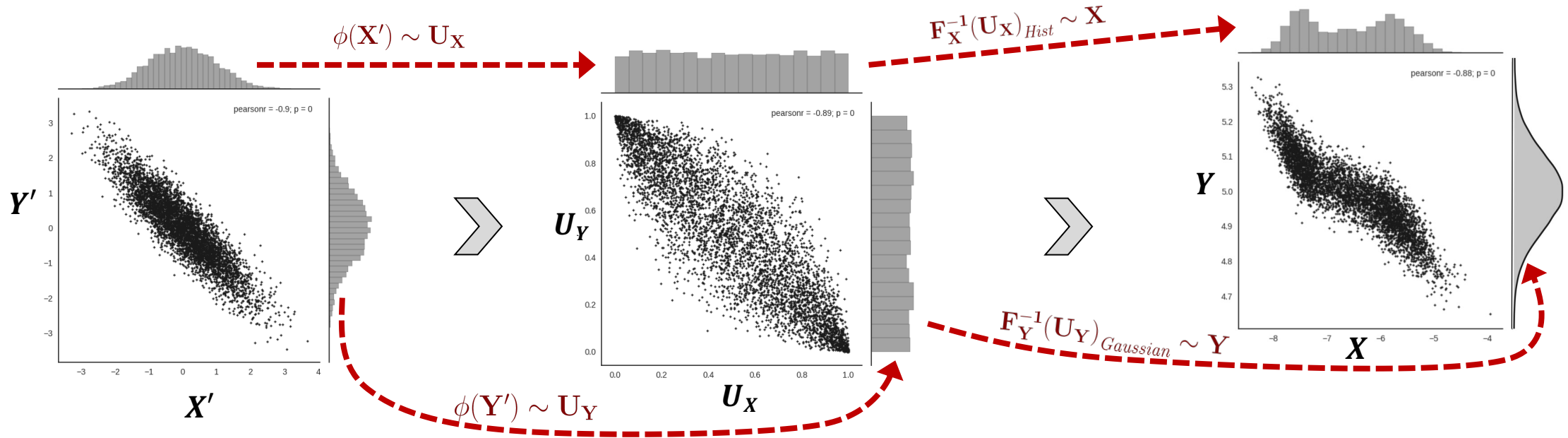
# Concept of Copula: Bivariate Example

- Consider a bivariate distribution of 2 random variables  $X$  and  $Y$  with a strong negative correlation coefficient of  $-0.9$
- Let,  $F_X$  and  $F_Y$  be their respective CDFs.
- So, we have:  $F_X$ ,  $F_Y$  and  $\rho = -0.9$



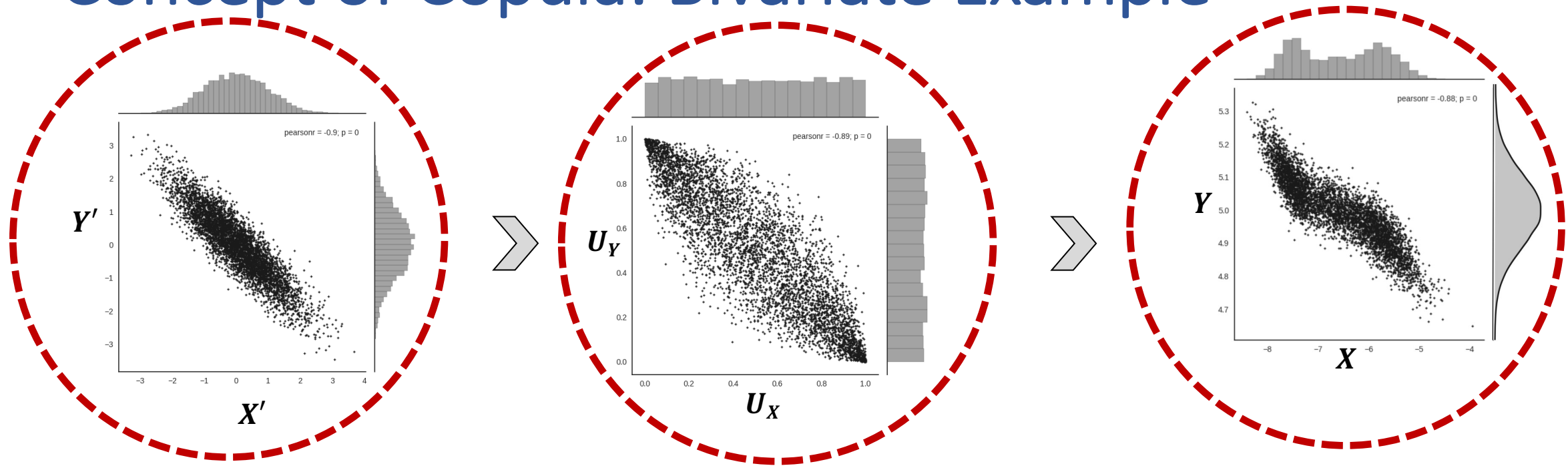


# Concept of Copula: Bivariate Example



- **Step 1:** Draw samples from a bivariate standard normal distribution with the desired correlation
- **Step 2:** Transform the Gaussian marginal distributions to Uniform distributions i.e.,  $U_X$  and  $U_Y$  **[Property 2]**
- **Step 3:** Transform  $U_X$  and  $U_Y$  to the desired marginal distribution forms using the inverse function of  $F_X$  and  $F_Y$  **[Property 1]**

# Concept of Copula: Bivariate Example



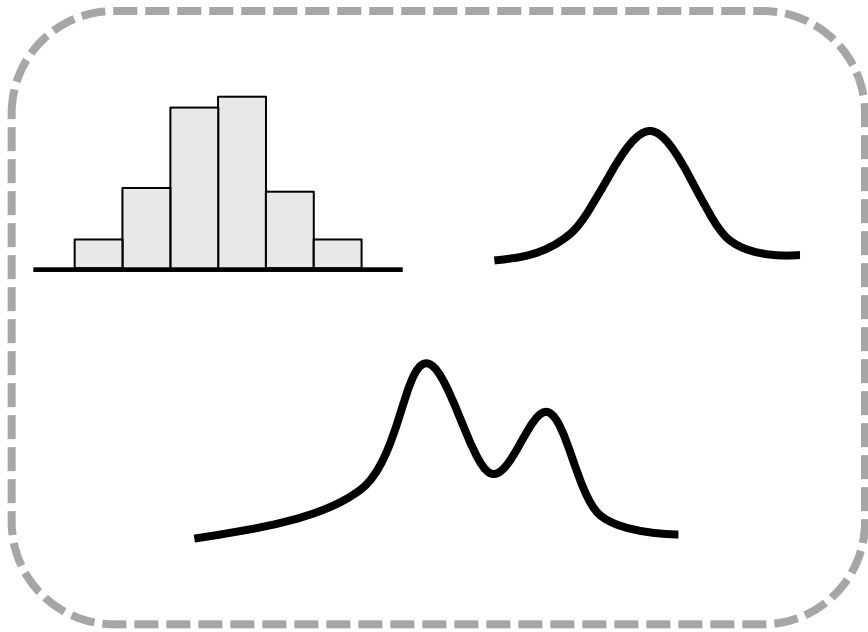
- Step 2 represents samples generated from a **Gaussian Copula**
- Gaussian Copula  $\rightarrow$  standard normal distribution
- Final distribution  $\rightarrow$  **Meta-Gaussian** distribution

# Multivariate Data Summary Using Copula

- What information to store in our multivariate data summaries?

# Multivariate Data Summary Using Copula

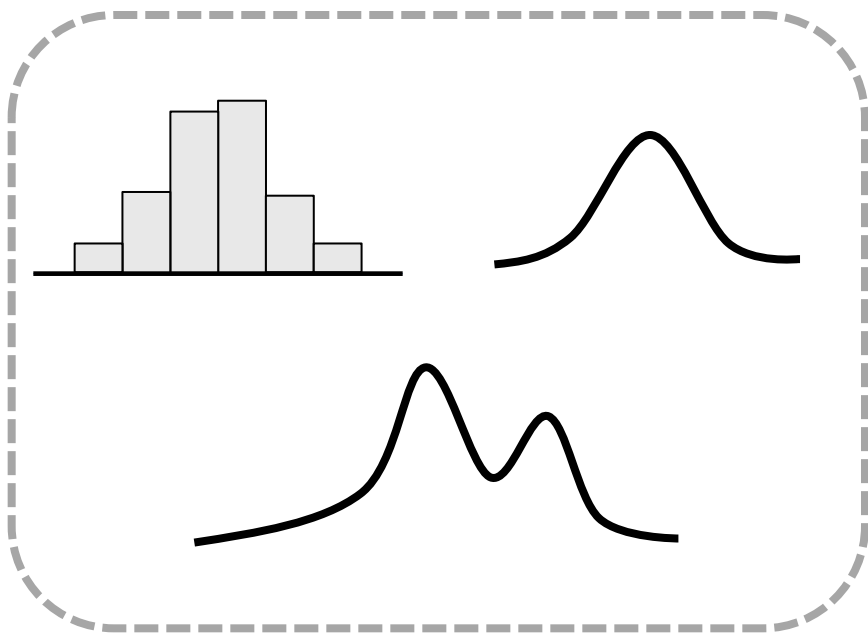
- Consider multivariate system of variables  $v_1, v_2, v_3, v_4$
- Univariate distributions,  $\theta_i \rightarrow$  distribution parameters for  $i^{\text{th}}$  variable
- Dependency structure (correlation matrix for Gaussian Copula)



$$\begin{matrix} & v_1 & v_2 & v_3 & v_4 \\ v_1 & \begin{pmatrix} 1 & \rho_{12} & \rho_{13} & \rho_{14} \\ \rho_{21} & 1 & \rho_{23} & \rho_{24} \\ \rho_{31} & \rho_{32} & 1 & \rho_{34} \\ \rho_{41} & \rho_{42} & \rho_{43} & 1 \end{pmatrix} \\ v_2 & \\ v_3 & \\ v_4 & \end{matrix}$$

# Multivariate Data Summary Using Copula

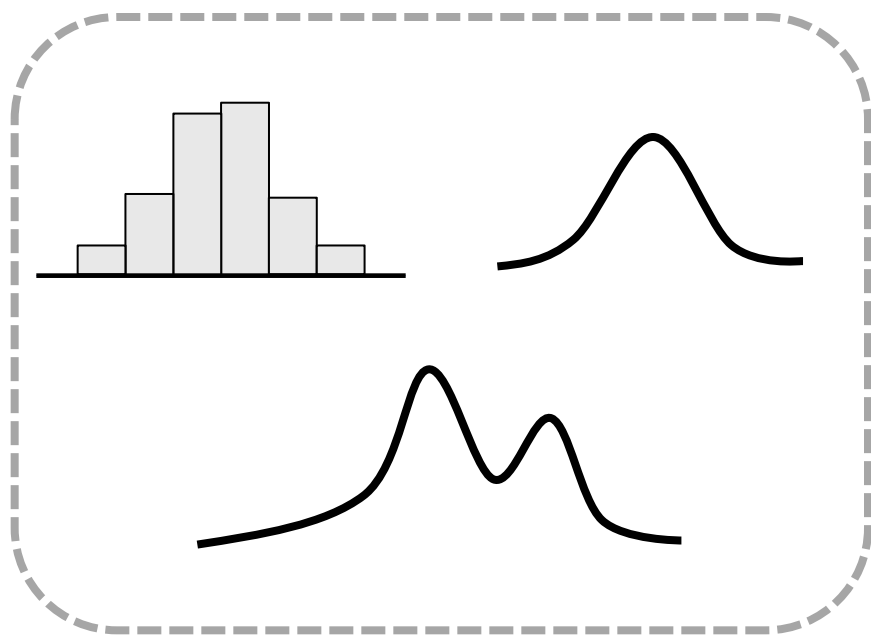
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# Multivariate Data Summary Using Copula

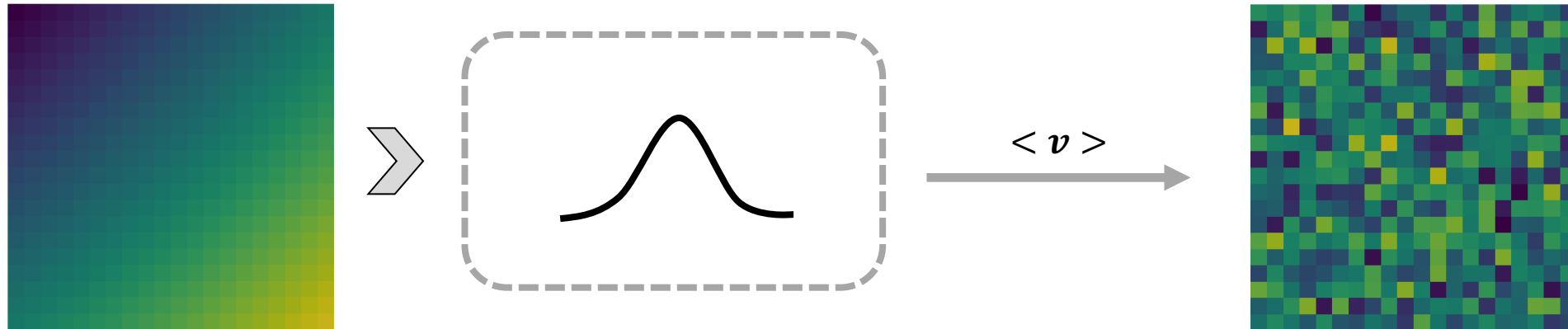
Storage footprint =  $\sum_{i=1}^n \text{size}(\theta_i) +$   
elems from upper (or lower)triangular matrix)



$$\begin{matrix} & \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 & \mathbf{v}_4 \\ \mathbf{v}_1 & \left( \begin{array}{cccc} 1 & \rho_{12} & \rho_{13} & \rho_{14} \\ \rho_{21} & 1 & \rho_{23} & \rho_{24} \\ \rho_{32} & \rho_{32} & 1 & \rho_{34} \\ \rho_{42} & \rho_{42} & \rho_{43} & 1 \end{array} \right) \\ \mathbf{v}_2 & \\ \mathbf{v}_3 & \\ \mathbf{v}_4 & \end{matrix}$$

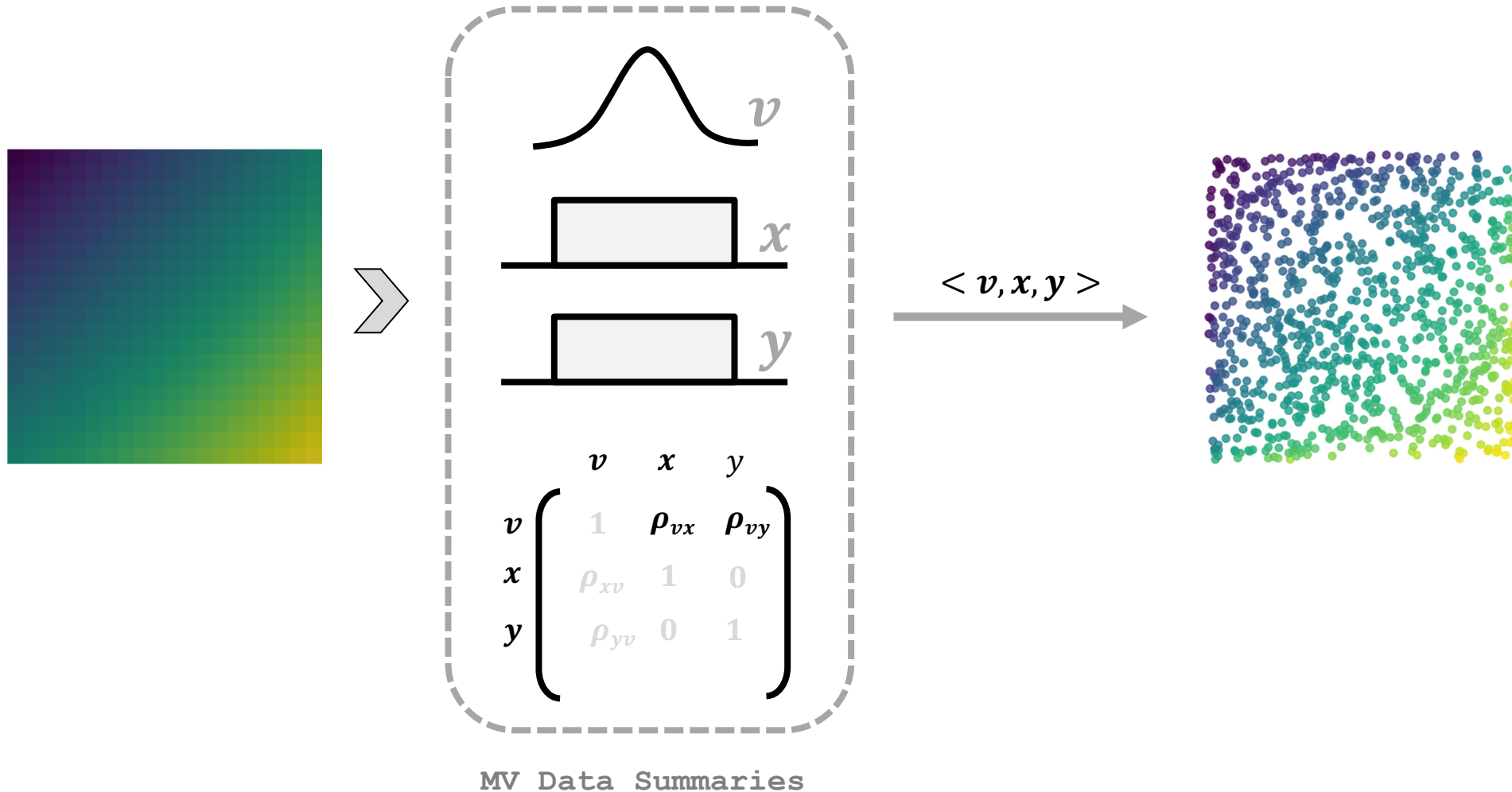
# Spatial Distributions

- Storing only the value distribution misses the spatial context of the data



# Spatial Distributions

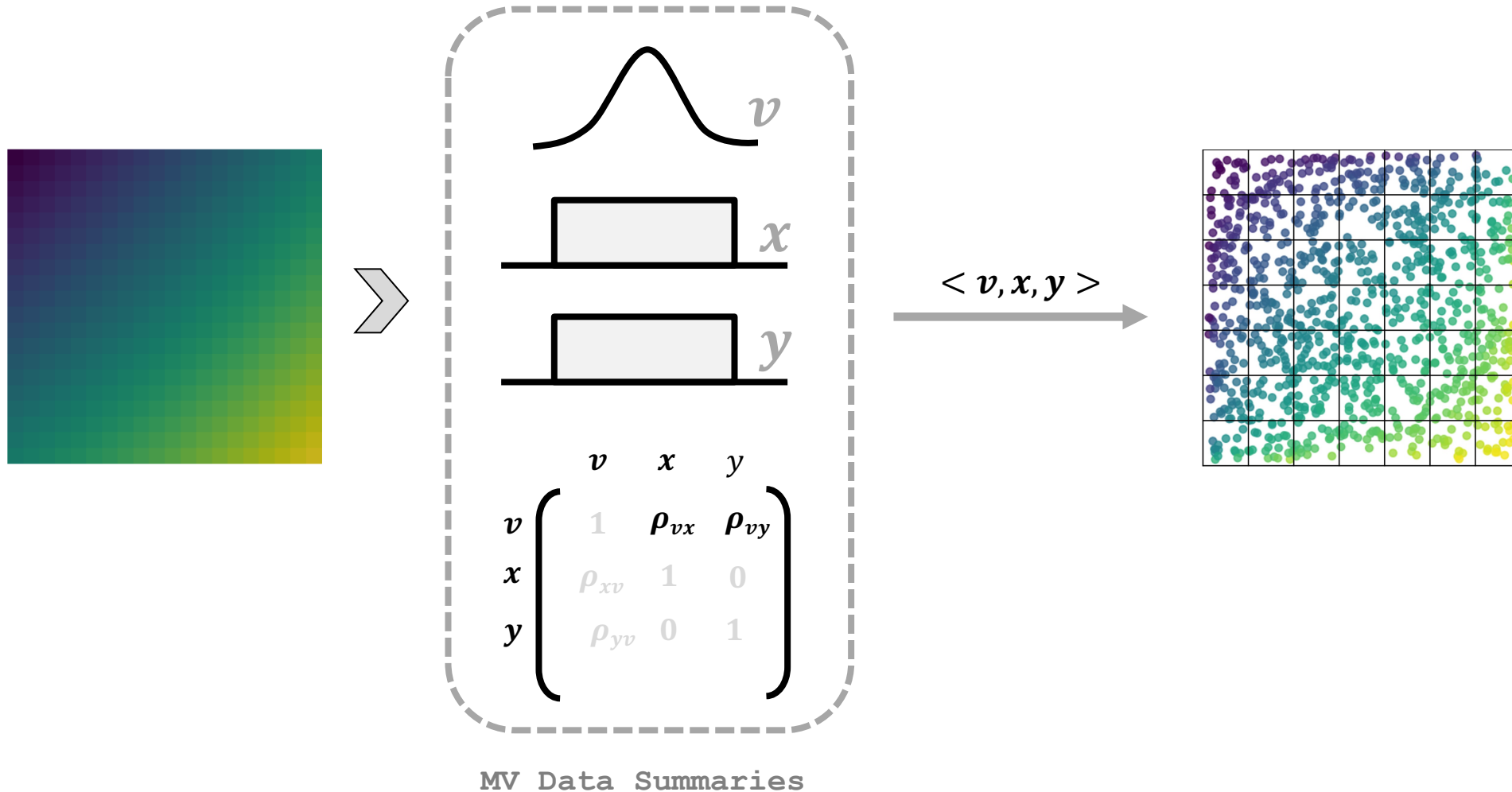
- Consider  $x$  and  $y$  distributions (uniform) along with  $v$





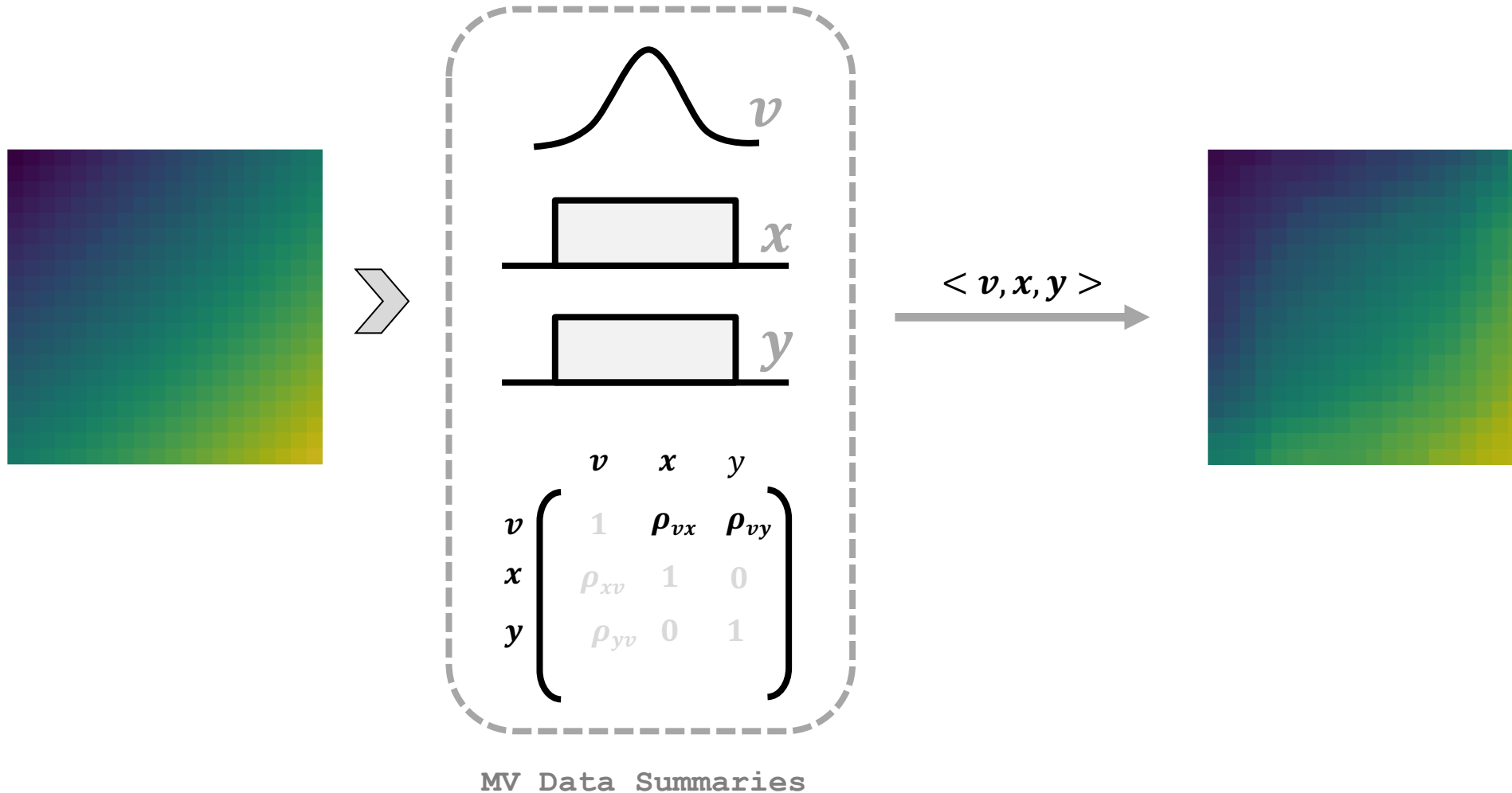
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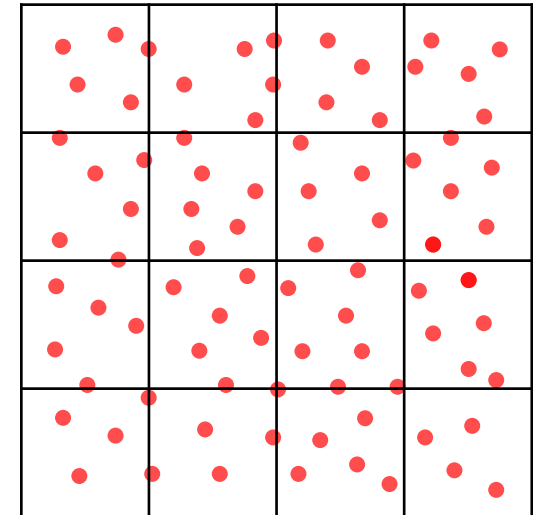
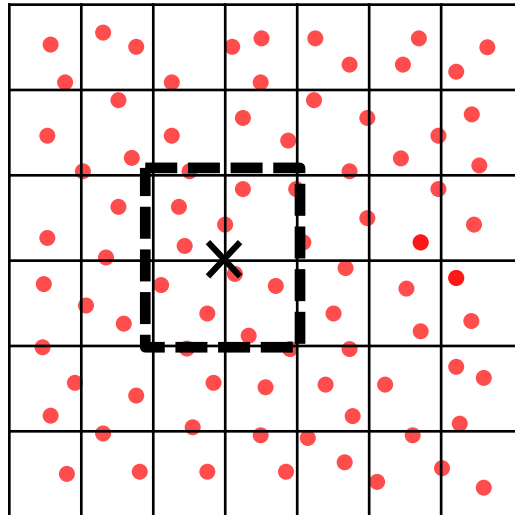
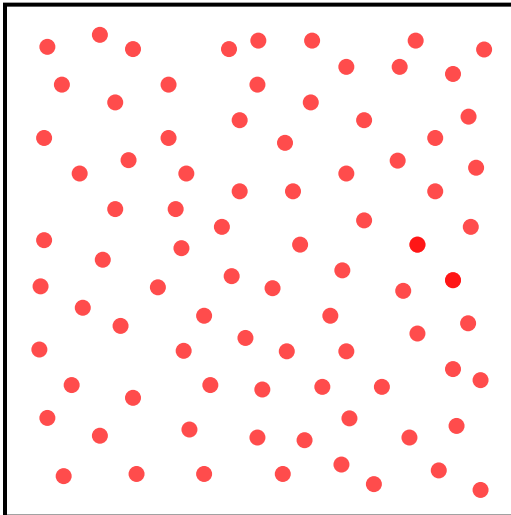
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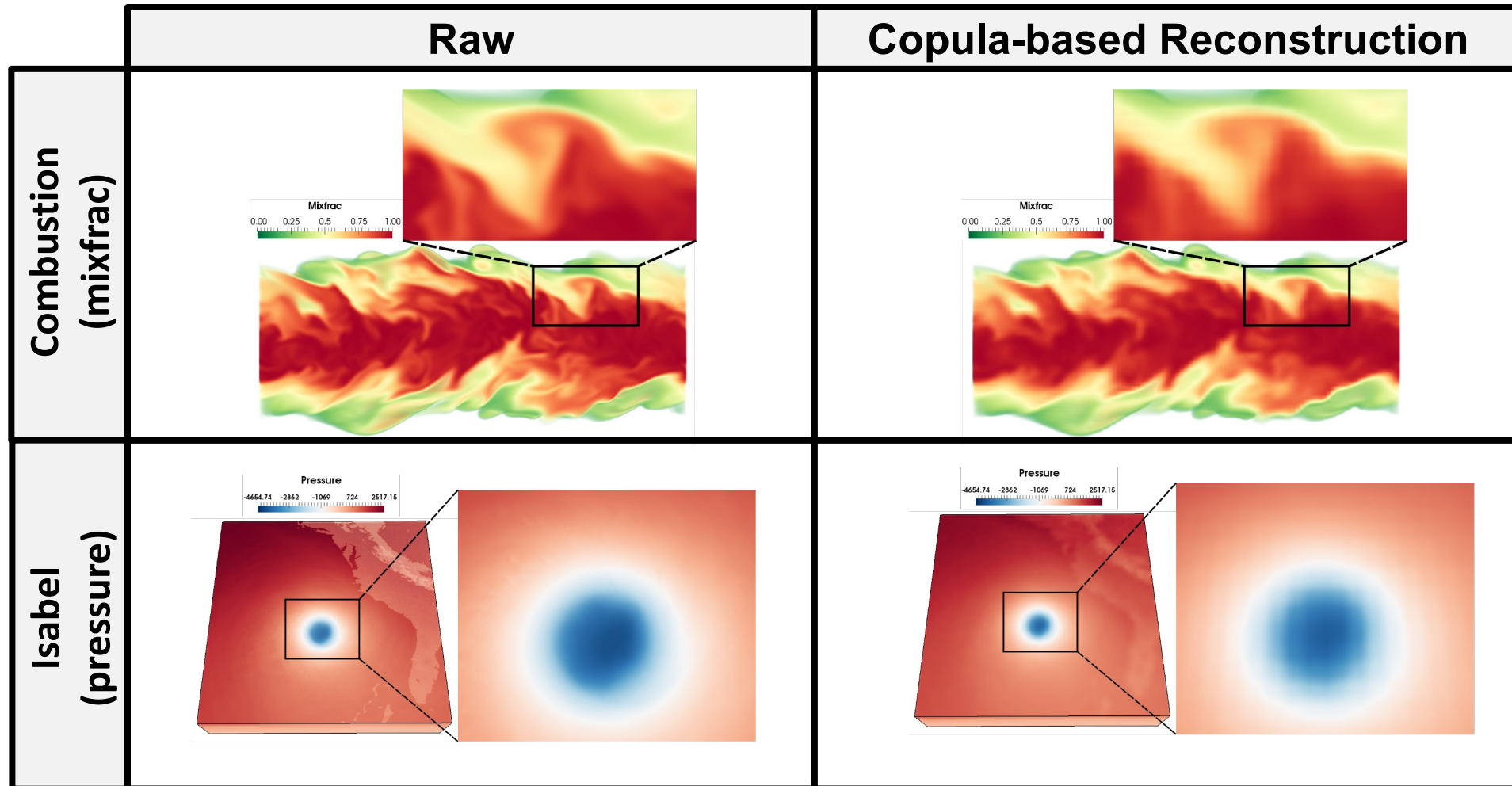


# Multivariate Sampling For Reconstruction

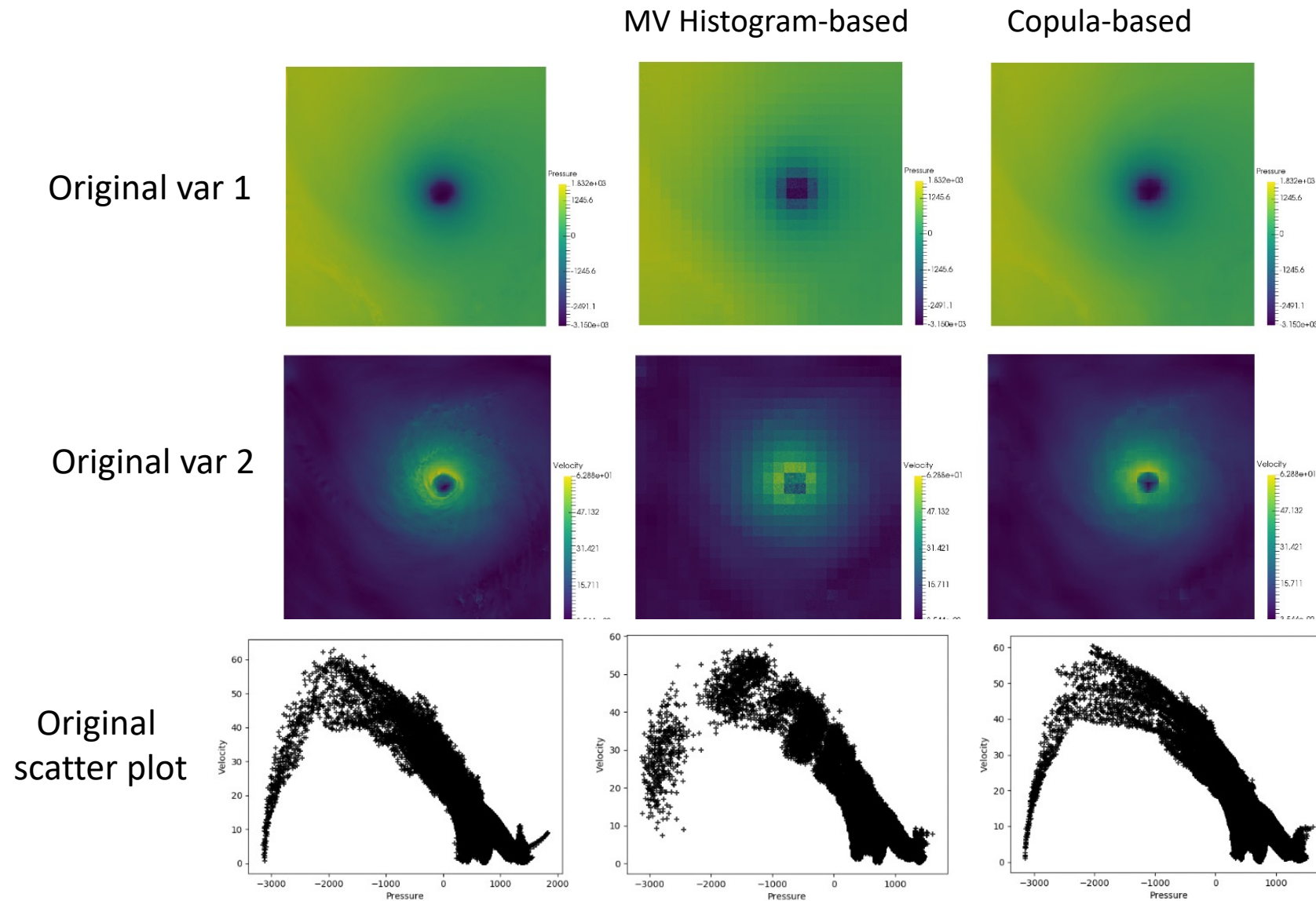
- Create statistical realizations of the scalar fields from the multivariate data summaries
- Utilize the spatial information to reconstruct the scalar field by computing local particle density estimate for the grid locations
- The sample scalar field can be generated in arbitrary user-specified grid resolution



# Multivariate Sampling-based Visualization

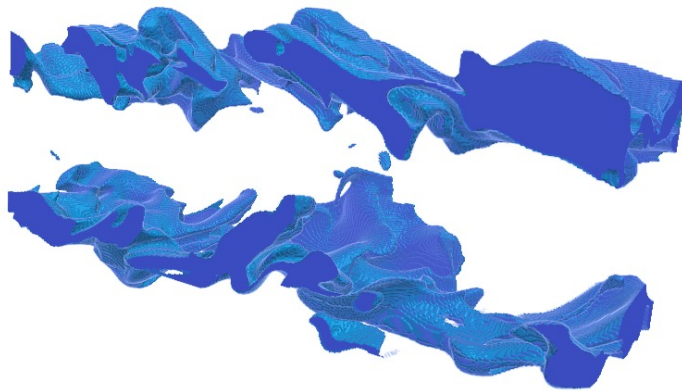


# Preserving Correlations

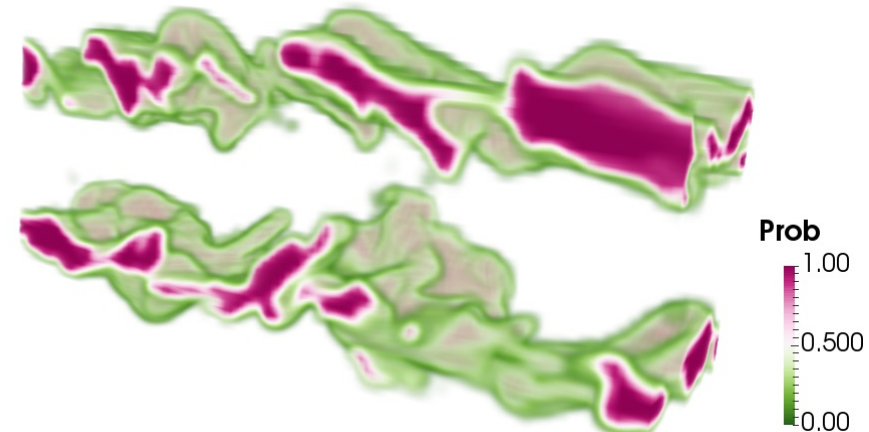


# Probabilistic Query-driven Analysis

- Query-driven methods are a class of highly effective visualization strategies
- Analysis tasks can be targeted only for the queried regions
- Selectively sample only from distributions which satisfy the user query



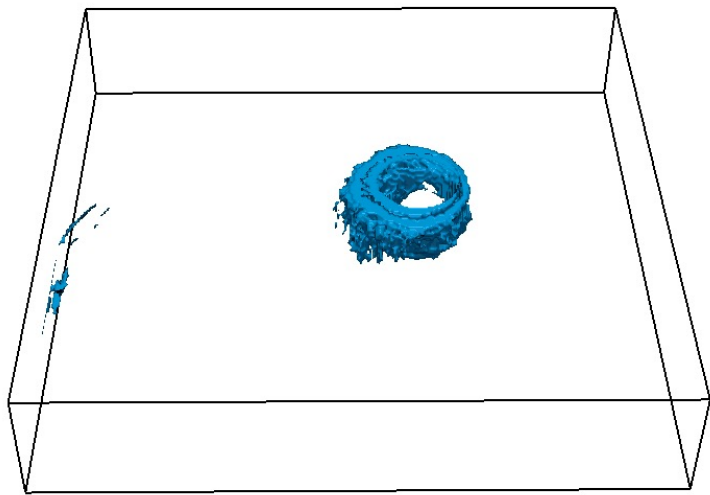
$0.3 < \text{Mixfrac} < 0.7 \text{ AND } y_{oh} > 0.0006$



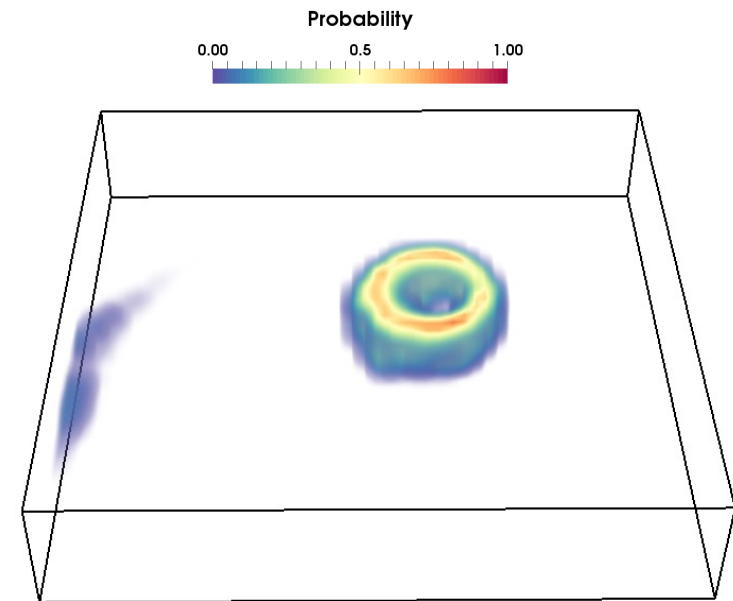
$P(0.3 < \text{Mixfrac} < 0.7 \text{ AND } y_{oh} > 0.0006)$

# Probabilistic Query-driven Analysis

- Query-driven methods are a class of highly effective visualization strategies
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$-2000 < \text{Pressure} < 500$  **AND**  $40 < \text{Velocity} < 50$



$P(-2000 < \text{Pressure} < 500 \text{ AND } 40 < \text{Velocity} < 50)$



# Performance and Storage

| Dataset<br>(Resolution)     | #variables | Raw<br>Size<br>(MB) | block<br>size | MV Histogram |                  | MV GMM       |                  | Hybrid<br>+ Copula |                  |
|-----------------------------|------------|---------------------|---------------|--------------|------------------|--------------|------------------|--------------------|------------------|
|                             |            |                     |               | Size<br>(MB) | Est.<br>Time (s) | Size<br>(MB) | Est.<br>Time (s) | Size<br>(MB)       | Est.<br>Time (s) |
| Isabel<br>(250x250x50)      | 11         | 137.5               | 5x5x5         | 173.1        | 106.1            | 23.7         | 2623.6           | 16.2               | 203.9            |
|                             |            |                     | 7x7x7         | 152.5        | 111.5            | 8.13         | 4671.6           | 5.8                | 205.4            |
|                             |            |                     | 10x10x10      | 113.7        | 98.2             | 2.95         | 5006.2           | 2.2                | 230.2            |
| Combustion<br>(480x720x120) | 3          | 497.7               | 5x5x5         | 579.4        | 311.7            | 55.7         | 4077.7           | 39.2               | 573.3            |
|                             |            |                     | 7x7x7         | 509.1        | 322.4            | 39.7         | 5150.4           | 14.3               | 561.7            |
|                             |            |                     | 10x10x10      | 434.2        | 305.7            | 27.8         | 9708.5           | 5.1                | 583.6            |

- Copula-based modeling + hybrid distribution-based summarization
  - 496.8 GB raw data could be reduced down to 19.8GB distribution-based data